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Kuniamuthur, Coimbatore – 641008
Phone : (0422)-2678001 (7 Lines) | Email : info@skcet.ac.in | Website : www.skcet.ac.in

AI WORKFLOW AGENTS TO SUPPORT TEACHERS IN UNDER RESOURCED, MULTI-GRADE LEARNING ENVIRONMENTS

A PROJECT REPORT

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Submitted by

KANIGA R A (727722EUAI026),

KANISH YATHRA RAJ R (727722EUAI027),

WILSONPAULRAJ D (727722EUAI070)

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degree of

BACHELOR OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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BONAFIDE CERTIFICATE

Certified that this project report **"AI WORKFLOW AGENTS TO SUPPORT TEACHERS IN UNDER-RESOURCED, MULTI-GRADE LEARNING ENVIRONMENTS"** is the bonafide work of "**KANIGA R A (727722EUAI026), KANISH YATHRA RAJ R (727722EUAI027), WILSONPAULRAJ D (727722EUAI070)"** who carried out the project work under my supervision.

SIGNATURE

Dr. K. VENKATA LAKSHMI M.TECH., PhD

HEAD OF THE DEPARTMENT
Professor

Department of Artificial Intelligence And Data Science

Sri Krishna College Of Engineering And Technology, Coimbatore-641008.

SIGNATURE

Mrs. RAASHMA MAHABOOB M.TECH.,

SUPERVISOR
Assistant Professor

Department of Artificial Intelligence And Data Science

Sri Krishna College Of Engineering And Technology, Coimbatore-641008.

Submitted for the Project viva-voce examination held on_____

INTERNAL EXAMINER

EXTERNAL EXAMINER

SUSTAINABLE DEVELOPMENT GOALS

The Sustainable Development Goals are a collection of 17 global goals designed to blue print to achieve a better and more sustainable future for all. The SDGs, set in 2015 by the United Nations General Assembly and intended to be achieved by the year 2030, In 2015, 195 nations agreed as a blue print that they can change the world for the better. The project is based on one of the 17 goals.

SDG	IMPLEMENTATION
SDG 4 (Quality Education)	This project enhances inclusive and equitable quality education by providing AI-powered teaching support to under-resourced and multi-grade classrooms. It automates lesson planning, content generation, and student performance tracking, reducing teacher workload and ensuring better learning outcomes. The system promotes personalized education and bridges the digital divide between rural and urban schools.
SDG 9 (Industry, Innovation, and Infrastructure)	This project fosters innovation in the education sector by integrating AI technologies such as Large Language Models (LLMs) for workflow automation. It strengthens digital infrastructure in schools, promotes sustainable EdTech development, and enables scalable, intelligent educational solutions that support teachers and learners in diverse environments.

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ABSTRACT

In today's educational landscape, teachers in under-resourced and multi-grade classrooms face significant challenges in managing time, preparing teaching materials, and tracking student performance effectively. This project presents a unified, AI-driven platform designed to serve as an intelligent teaching assistant that automates the complete workflow of educators. The system leverages Large Language Models (LLMs) and machine learning algorithms to reduce manual effort and streamline core teaching processes such as timetable generation, lecture preparation, and performance evaluation.

Teachers begin by entering their subjects, grades, and syllabus details into the platform. Based on this input, the system automatically generates a personalized, conflict-free timetable and displays it through an interactive calendar view. Before each class, the AI module prepares context-aware teaching materials—including lecture notes, PowerPoint slides, quizzes, and assessments—ensuring that the content aligns precisely with the curriculum and student level. When classes are missed or topics remain incomplete, the system intelligently reschedules them based on feedback and availability.

The platform also supports seamless integration with existing educational ecosystems such as Google Classroom, enabling teachers to share AI-generated content and assignments effortlessly. After assessments, teachers can upload student marks, which are then analyzed to produce visual insights and trend graphs that highlight learning progress and identify knowledge gaps. This data-driven approach empowers teachers to adopt more personalized teaching strategies and helps students receive the right academic support at the right time.

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LIST OF ABBREVIATIONS

ACRONYM	ABBREVIATIONS
AI	ARTIFICIAL INTELLIGENCE
LLM	LARGE LANGUAGE MODEL
UI	USER INTERFACE
UX	USER EXPERIENCE
API	APPLICATION PROGRAMMING INTERFACE
ML	MACHINE LEARNING
NLP	NATURAL LANGUAGE PROCESSING
CSV	COMMA-SEPARATED VALUES

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